

Job sequencing

Let there ~~be~~ be 14 jobs with the profit of 22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 27, 19, 11.

Job deadlines 3, 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 2, 14

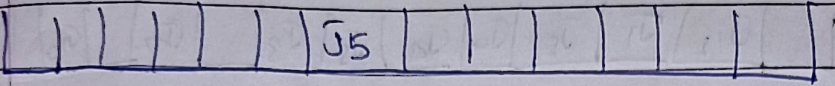
Job Sequencing Solution

S1: Arrange the jobs in descending order of profit

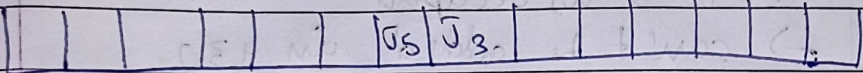
Job	Profit	Deadline
J5	30	7
J3	29	8
J4	28	6
J7	27	10
J12	27	2
J10	26	12
J8	25	4
J9	24	6
J1	22	3
J6	21	5
J2	19	3
J13	19	14
J11	14	13
J14	11	1

S2: assign jobs at optimal time slots.

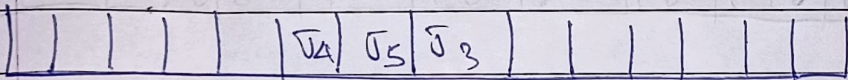
1) add J_5 (30, 7)



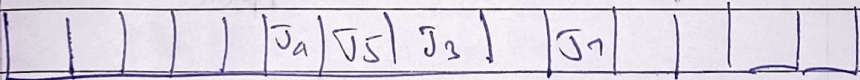
2) add J_3 (29, 8)



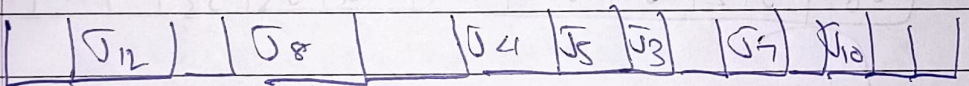
3) add J_4 (28, 6)



4) add J_7 (27, 10)



5) J_{12} (27, 2), J_{10} (26, 12), J_8 (25, 4)

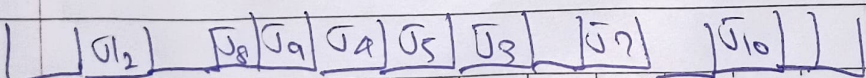


Sequence: $J_5, J_3, J_4, J_7, J_{12}, J_{10}, J_8$

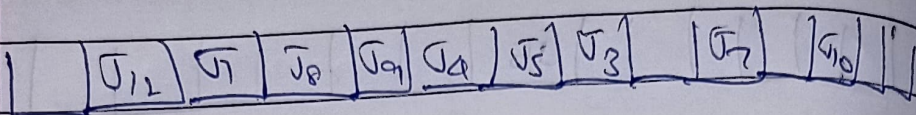
6) Add J_9 (24, 6)

→ J_9 can't be added into slot 6, since it is already occupied

→ It can occupy the slot ⁵ ~~6, since it is already~~



⑦ $J_1 (22, 3)$

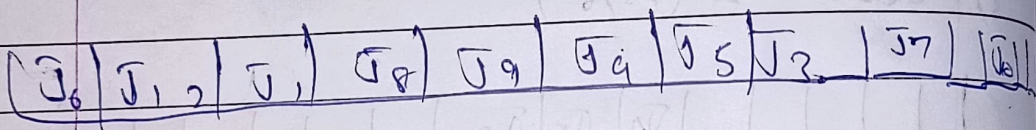


⑧ Add $J_6 (21, 5)$

→ slot 5 is occupied

→ can't be added in 4, 3

→ can be added in slot 1

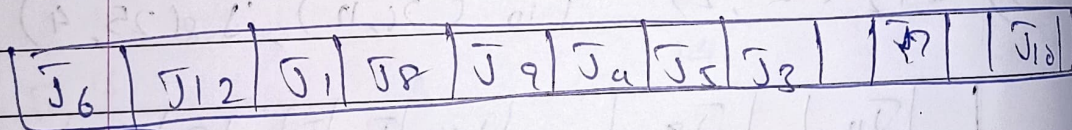


⑨ Add $J_2 (19, 3)$

→ J_2 in slot 3, 2, 1

so it can be skipped

⑩ $J_{13} (19, 14)$, $J_{11} (14, 13)$



sequence = $\{J_5, J_3, J_4, J_7, J_2, J_{10}, J_8, J_9, J_1, J_6, J_{13}, J_{11}\}$

⑪ $J_{14} (11, 1)$

→ already occupied

final job sequence